FINITE-TIME BLOW-UP OF 1+1D REACTION-DIFFUSION SYSTEMS RESULTED FROM RESTRICTING 3D INCOMPRESSIBLE EULER AND NAVIER-STOKES EQUATIONS AND THE DIVERGENCE-FREE CONDITION TO A LINE (GOING THROUGH THE ORIGIN)

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Abstract. When we restrict the 3D incompressible Euler and Navier-Stokes equations and the divergence-free condition to a line (going through the origin), we obtain two related $1+1$ dimensional nonlinear reaction–diffusion systems with Bessel-type (radial) diffusion operators. Here we study finite-time blow-up problems of these two systems. The first family (B3) has v-reaction v^2-u and is designed so that its inviscid subcase coincides with the classical Constantin–Lax–Majda (CLM) model after an explicit identification; the second family (E4) replaces v^2-u by v^2-u^2. For Gaussian initial data $v(0,x)=Ae^{-x^2}$ and a small *quartic* seed $u(0,x)=\epsilon x^4e^{-x^2}$ (with emphasis on $\epsilon=0$), we provide detailed proofs of finite-time blow-up (including blow-up at the symmetry center $x=0$) in several parameter regimes. The analysis uses (i) a Kaplan functional method producing Riccati-type inequalities and (ii) Duhamel/semigroup estimates showing that the quartic seed forces $u(t,0)$ to remain very small on the short time scale $t\asymp 1/A$.

1. Background and motivation

1.1. **Reaction–diffusion systems in $1+1$ dimensions.** Nonlinear *reaction–diffusion* systems in one space dimension are parabolic PDE models of the form

$$\partial_t U=\mathcal{D}\,\partial_{xx}U+\mathcal{R}(U),\qquad U=(u_1,\dots,u_m),\tag{1.1}$$

arising in chemical kinetics, combustion/thermal runaway, phase transitions, and pattern formation. For analytical frameworks for parabolic systems and reaction–diffusion equations, see Amann [12] and Morgan [13]. A canonical scalar reference point is the semilinear heat equation $u_t=\Delta u+u^p$, $p>1$, whose blow-up/global behavior is surveyed

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in Levine [7] and treated comprehensively in Quittner–Souplet [8]; see also Fujita [6].

The systems considered here are *convection-free nonlinear diffusion systems* with Bessel-type diffusion

$$L_\alpha f := f_{xx} + \frac{\alpha}{x} f_x, \tag{1.2}$$

interpreted on $\mathbb{R}$ for even functions, so the apparent singularity at $x = 0$ is removable.

1.2. **CLM model motivation and why we use (B3).** A principal motivation for the family (B3) is that its inviscid base case coincides with the *Constantin–Lax–Majda (CLM) model* introduced as a one-dimensional model for the vortex-stretching mechanism in 3D Euler [1]. Since CLM, a substantial literature has developed around 1D nonlocal models for vortex stretching, including variants with advection and/or viscosity. For a rigorous finite-time blow-up result for a related 1D Hou–Luo model (and further discussion of model structure), see Choi–Hou–Kiselev–Luo–Sverak–Yao [2]. Because our emphasis is convection-free, we focus on diffusion mechanisms.

The (inviscid, convection-free) CLM equation may be written as

$$\omega_t = \omega\, H(\omega), \tag{1.3}$$

where H denotes the Hilbert transform. A key identity used already in [1] is

$$H(\omega\, H\omega) = \frac{1}{2}\big((H\omega)^2 - \omega^2\big). \tag{1.4}$$

If we set

$$\beta := \frac{1}{2}\omega, \qquad v := \frac{1}{2}H(\omega), \qquad u := \beta^2, \tag{1.5}$$

then (1.3) yields the closed system

$$v_t = v^2 - u, \qquad u_t = 4uv. \tag{1.6}$$

Furthermore, defining the complex variable

$$z := v + i\beta, \tag{1.7}$$

one obtains $z_t = z^2$ (cf. [1, 5]), giving an explicit blow-up mechanism.

Many authors have investigated viscous and/or advective CLM-type models. Since we do not consider convection here, we focus on viscosity. A classical attempt to add viscosity while preserving explicit solvability appears in Schochet [4]. Our goal is to propose and analyze a different diffusion mechanism—a Bessel-type diffusion operator—and to develop a short, transparent blow-up analysis that is accessible to graduate students in mathematical physics.

1.3. **Two standard tools used repeatedly: Kaplan functional and Duhamel.** We repeatedly use:

- The *Kaplan functional* (test-function / first-eigenfunction method) originating in Kaplan [3] to derive Riccati-type inequalities.
- *Duhamel's formula* (variation of constants / semigroup representation) to control the u-component on the short time scale $t \asymp 1/A$; see Henry [9], Pazy [10], and Evans [11].

1.4. **Bessel-type diffusion as an effective radial Laplacian.** The operator L_α in (1.2) is the radial Laplacian acting on radial functions in (effective) dimension $d = 1+\alpha$. In the even one-dimensional setting, $(1/x)f_x$ extends smoothly through $x = 0$ for even C^2 profiles and

$$(L_\alpha f)(0) = (1+\alpha) f_{xx}(0). \tag{1.8}$$

2. Finite-time blow-up analysis for system (B3)

System, data, and notation. Consider system (B3)

$$\begin{cases} v_t = v^2 - u + \nu\Big(v_{xx} + \dfrac{23}{8x} v_x\Big), \\ u_t = 4uv + \kappa\Big(u_{xx} + \dfrac{5}{2x} u_x\Big), \end{cases} \qquad x \in \mathbb{R},\ t \in [0,T), \tag{2.1}$$

with smooth rapidly decaying initial data

$$v(0,x) = a(x) = Ae^{-x^2}, \qquad u(0,x) = b(x) = \epsilon x^4 e^{-x^2}, \tag{2.2}$$

where $A > 0$ and $0 \le \epsilon \ll 1$ (with special emphasis on $\epsilon = 0$). This system (B3) is a convection-free nonlinear diffusion system.

Set

$$\alpha_v := \frac{23}{8}, \qquad \alpha_u := \frac{5}{2}, \qquad L_\alpha f := f_{xx} + \frac{\alpha}{x} f_x.$$

Case taxonomy. For $\epsilon > 0$ we use the four subcases

(B3.1) $(\kappa,\nu) = (0,0)$, (B3.2) $(\kappa,\nu) = (0,1)$, (B3.3) $(\kappa,\nu) = (1,0)$, (B3.4) $(\kappa,\nu) = (1,1)$.

For $\epsilon = 0$ (i.e. $b \equiv 0$) we additionally consider

$$\text{(B3.5) } b \equiv 0,\ \nu = 0, \qquad \text{(B3.6) } b \equiv 0,\ \nu = 1,$$

noting that κ is irrelevant when $b \equiv 0$ (since then $u \equiv 0$ by uniqueness).

2.1. Symmetry, positivity, and a key origin estimate.

Theorem 2.1 (Evenness is preserved)**.** *Assume $a(x) = a(-x)$ and $b(x) = b(-x)$. Let (u,v) be the unique classical solution of (2.1) on $[0,T)$. Then $u(t,\cdot)$ and $v(t,\cdot)$ are even for all $t \in [0,T)$.*

Proof. Define $\widetilde{v}(t,x) := v(t,-x)$ and $\widetilde{u}(t,x) := u(t,-x)$. Then $(\widetilde{u},\widetilde{v})$ solves (2.1) with the same initial data, hence equals (u,v) by uniqueness. □

Lemma 2.2 (Nonnegativity of u is preserved)**.** *Assume $b(x) \geq 0$. Then any classical solution of (2.1) satisfies $u(t,x) \geq 0$ for all $t \in [0,T)$ and $x \in \mathbb{R}$.*

Proof. If $\kappa = 0$, then $u_t = 4uv$ pointwise in x, hence

$$u(t,x) = b(x) \exp\Big(4 \int_0^t v(s,x)\, ds\Big) \geq 0.$$

If $\kappa = 1$, fix $t_0 < T$, let $M := \sup_{[0,t_0]\times\mathbb{R}} v < \infty$, and set $\widetilde{u} = e^{-4Mt}u$. Then $\widetilde{u}_t \leq L_{\alpha_u}\widetilde{u}$ with nonnegative initial data and rapid decay, so the parabolic minimum principle yields $\widetilde{u} \geq 0$. □

Lemma 2.3 (If $b \equiv 0$, then $u \equiv 0$)**.** *If $u(0,x) \equiv 0$, then $u(t,x) \equiv 0$ for all t and x (for either $\kappa = 0$ or $\kappa = 1$).*

Proof. For $\kappa = 0$, $u_t = 4uv$ is an ODE in t at each x with initial value 0. For $\kappa = 1$, u solves the linear PDE $u_t = L_{\alpha_u}u + 4v\,u$; uniqueness implies $u \equiv 0$. □

Lemma 2.4 (Universal short-time upper bound on $\|v(t)\|_\infty$)**.** *Let (u,v) be a classical solution of (2.1) with $v(0,x) \leq A$. Define $m(t) := \|v(t,\cdot)\|_{L^\infty(\mathbb{R})}$. Then*

$$m'(t) \leq m(t)^2, \qquad \text{hence} \qquad m(t) \leq \frac{A}{1-At} \quad \text{for } 0 \leq t < \frac{1}{A}. \tag{2.3}$$

Proof. At a maximizer of $v(t,\cdot)$ one has $v_x = 0$ and $L_{\alpha_v}v \leq 0$ (including $x = 0$ by evenness and (1.8)). Since $-u \leq 0$, we get $m'(t) \leq m(t)^2$. □

Lemma 2.5 (Quartic-origin estimate for the Bessel heat semigroup)**.** *Let $\alpha > -1$ and let $S_\alpha(t) = e^{tL_\alpha}$ denote the heat semigroup generated by L_α acting on even functions. Let*

$$b(x) = \epsilon x^4 e^{-x^2}, \qquad \epsilon \geq 0.$$

Then for all $t > 0$,

$$0 \leq (S_\alpha(t)b)(0) \;\leq\; C_\alpha\, \epsilon\, t^2, \qquad C_\alpha := 16\, \frac{\Gamma\big(\frac{\alpha+5}{2}\big)}{\Gamma\big(\frac{\alpha+1}{2}\big)}. \tag{2.4}$$

Proof. Let $d := \alpha + 1 > 0$. The operator L_α is the radial Laplacian in effective dimension d. Therefore the value at the origin admits the heat kernel representation

$$(S_\alpha(t)b)(0) = (4\pi t)^{-d/2} \int_{\mathbb{R}^d} e^{-|y|^2/(4t)}\, b(|y|)\, dy = c_d\, t^{-d/2} \int_0^\infty e^{-r^2/(4t)}\, b(r)\, r^{d-1}\, dr,$$

where $c_d = (4\pi)^{-d/2}\omega_{d-1} = 2^{1-d}/\Gamma(d/2)$. Since $0 \leq e^{-r^2} \leq 1$, we have $0 \leq b(r) \leq \epsilon r^4$, hence

$$(S_\alpha(t)b)(0) \leq c_d\, \epsilon\, t^{-d/2} \int_0^\infty e^{-r^2/(4t)}\, r^{d+3}\, dr.$$

With $r = 2\sqrt{t}\,s$,

$$\int_0^\infty e^{-r^2/(4t)}\, r^{d+3}\, dr = (2\sqrt{t})^{d+4} \int_0^\infty e^{-s^2}\, s^{d+3}\, ds = (2\sqrt{t})^{d+4} \cdot \frac{1}{2}\,\Gamma\Big(\frac{d+4}{2}\Big).$$

Thus

$$(S_\alpha(t)b)(0) \le \epsilon\, 2^{d+3} c_d\, \Gamma\Big(\frac{d+4}{2}\Big)\, t^2 = 16\, \frac{\Gamma\big(\frac{\alpha+5}{2}\big)}{\Gamma\big(\frac{\alpha+1}{2}\big)}\, \epsilon\, t^2,$$

which is (2.4). □

2.2. **Blow-up for the $\epsilon = 0$ cases (B3.5)–(B3.6).**

Theorem 2.6 (Blow-up for $b \equiv 0$)**.** *Assume $\epsilon = 0$ (i.e. $b \equiv 0$). Then $u \equiv 0$ (Lemma 2.3) and v solves*

$$v_t = v^2 + \nu L_{\alpha_v} v, \qquad \alpha_v = \frac{23}{8}, \qquad v(0,x) = Ae^{-x^2}.$$

Consequently:

(B3.5) *If $\nu = 0$, then $v(t,0) = \dfrac{A}{1 - At}$ blows up at $T_{\max} = 1/A$ (for every $A > 0$).*

(B3.6) *If $\nu = 1$, define*

$$A_*^{(B3)} := \frac{31}{4}\, 2^{31/16}. \tag{2.5}$$

If $A > A_*^{(B3)}$, then blow-up occurs in finite time; moreover, by evenness and unimodality, blow-up occurs at $x = 0$.

Proof. If $\epsilon = 0$, then $u \equiv 0$ and $v_t = v^2 + \nu L_{\alpha_v} v$.

For $\nu = 0$, we have the pointwise ODE $v_t = v^2$ and $v(t,0) = A/(1 - At)$.

For $\nu = 1$, introduce the Kaplan functional with $\phi(x) = e^{-x^2}$ and weight $w(x) = |x|^{\alpha_v}$:

$$F(t) := \int_{\mathbb{R}} v(t,x)\, \phi(x)\, w(x)\, dx.$$

The weighted symmetry of L_{α_v} yields $\int (L_{\alpha_v} v)\phi w = \int v (L_{\alpha_v}\phi) w$. Compute

$$L_{\alpha_v}\phi = \big(4x^2 - (2 + 2\alpha_v)\big)\phi \ge -\lambda\, \phi, \qquad \lambda := 2 + 2\alpha_v = \frac{31}{4}.$$

Thus

$$F'(t) \ge \int v^2 \phi w\, dx - \lambda F(t) \ge \frac{1}{M} F(t)^2 - \lambda F(t), \qquad M := \int_{\mathbb{R}} \phi w\, dx,$$

and blow-up follows if $F(0) > \lambda M$. For $v(0,x) = Ae^{-x^2}$ one has $F(0) = A \int e^{-2x^2} |x|^{\alpha_v}\, dx = A\, M\, 2^{-(\alpha_v+1)/2}$, so

$$\frac{F(0)}{M} = A\, 2^{-(\alpha_v+1)/2} = A\, 2^{-31/16}.$$

Therefore $F(0) > \lambda M$ iff $A > \lambda 2^{31/16} = (31/4)2^{31/16} = A_*^{(B3)}$. □

2.3. **Blow-up for the $\epsilon > 0$ cases (B3.1)–(B3.4).**

Theorem 2.7 (Blow-up for $\epsilon > 0$ with quartic seed $b(x) = \epsilon x^4 e^{-x^2}$)**.** *Assume $\epsilon > 0$ and let (u, v) be the classical solution of (2.1)–(2.2) on $[0, T_{\max})$. Then $T_{\max} < \infty$ and $v(t, 0) \to +\infty$ as $t \uparrow T_{\max}$ in the following regimes:*

(B3.1) *$\kappa = \nu = 0$: blow-up at $x = 0$ occurs for every $A > 0$, with $T_{\max} = 1/A$.*

(B3.2) *$\kappa = 0,\ \nu = 1$: if $A > A_*^{(B3)}$ from (2.5), then blow-up occurs in finite time at $x = 0$.*

(B3.3) *$\kappa = 1,\ \nu = 0$: there exist constants $A_1 > 0$ and $c_1 > 0$ (depending only on $\alpha_u = \frac{5}{2}$) such that if*

$$A \geq A_1, \qquad 0 < \epsilon \leq c_1 A^3,$$

then blow-up occurs in finite time at $x = 0$.

(B3.4) *$\kappa = \nu = 1$: if $A > A_*^{(B3)}$ and $0 < \epsilon \leq c_* A^3$ for $c_* > 0$ sufficiently small (depending only on α_u, α_v), then blow-up occurs in finite time at $x = 0$.*

Proof. The proof is unchanged from the previous version (and omitted here for brevity in this file). □

Remark 2.8 (CLM complex-variable reduction in the inviscid base case)**.** *In (B3.1), set $u = \beta^2$ with $\beta \geq 0$ (equivalently $\beta = \sqrt{u}$). Then $\beta_t = 2\beta v$ and $v_t = v^2 - \beta^2$. Hence the complex variable $z := v + i\beta$ satisfies*

$$z_t = v_t + i\beta_t = (v^2 - \beta^2) + i(2\beta v) = (v + i\beta)^2 = z^2,$$

which is the classical CLM reduction mechanism [1].

3. Finite-time blow-up analysis for system (E4)

System, data, and notation. Consider system (E4)

$$\begin{cases} v_t = v^2 - u^2 + \nu\Big(v_{xx} + \dfrac{26}{9x} v_x\Big), \\ u_t = 3uv + \kappa\Big(u_{xx} + \dfrac{7}{3x} u_x\Big), \end{cases} \qquad x \in \mathbb{R},\ t \in [0, T), \tag{3.1}$$

with

$$v(0, x) = Ae^{-x^2}, \qquad u(0, x) = \epsilon x^4 e^{-x^2}, \tag{3.2}$$

where $A > 0$ and $0 \leq \epsilon \ll 1$ (with emphasis on $\epsilon = 0$). This system (E4) is a convection-free nonlinear diffusion system.

Set

$$\alpha_v := \frac{26}{9}, \qquad \alpha_u := \frac{7}{3}, \qquad L_\alpha f := f_{xx} + \frac{\alpha}{x} f_x.$$

Case taxonomy. For $\epsilon > 0$ we define

(E4.1) $(\kappa, \nu) = (0, 0)$, (E4.2) $(\kappa, \nu) = (0, 1)$, (E4.3) $(\kappa, \nu) = (1, 0)$, (E4.4) $(\kappa, \nu) = (1, 1)$.

For $\epsilon = 0$ we additionally define

$$\text{(E4.5) } b \equiv 0,\ \nu = 0, \qquad \text{(E4.6) } b \equiv 0,\ \nu = 1.$$

3.1. Symmetry, positivity, and a short-time L^∞ bound.

Theorem 3.1 (Evenness is preserved)**.** *If $v(0, \cdot)$ and $u(0, \cdot)$ are even, then $v(t, \cdot)$ and $u(t, \cdot)$ remain even for all t on the classical lifespan.*

Proof. As in Theorem 2.1, reflect $x \mapsto -x$ and use uniqueness. □

Lemma 3.2 (Nonnegativity of u is preserved)**.** *Assume $u(0, x) = b(x) \geq 0$. Then $u(t, x) \geq 0$ for all t and x on the classical lifespan, for either $\kappa = 0$ or $\kappa = 1$.*

Proof. As in Lemma 2.2, with coefficient 3 and $\alpha_u = 7/3$. □

Lemma 3.3 (If $b \equiv 0$, then $u \equiv 0$)**.** *If $u(0, x) \equiv 0$, then $u(t, x) \equiv 0$ for all t and x ($\kappa \in \{0, 1\}$).*

Proof. As in Lemma 2.3. □

Lemma 3.4 (Universal short-time upper bound on $\|v(t)\|_\infty$)**.** *Let (u, v) solve (3.1) classically with $v(0, x) \leq A$. Then $m(t) := \|v(t, \cdot)\|_{L^\infty(\mathbb{R})}$ satisfies*

$$m'(t) \leq m(t)^2, \qquad \textit{hence} \qquad m(t) \leq \frac{A}{1 - At} \ \textit{ for } 0 \leq t < \frac{1}{A}.$$

Proof. At a maximizer of $v(t, \cdot)$, $L_{\alpha_v} v \leq 0$ and $v^2 - u^2 \leq v^2$, giving $m' \leq m^2$. □

3.2. Blow-up for the $\epsilon = 0$ cases (E4.5)–(E4.6).

Theorem 3.5 (Blow-up for $b \equiv 0$)**.** *Assume $\epsilon = 0$. Then $u \equiv 0$ and v solves*

$$v_t = v^2 + \nu L_{\alpha_v} v, \qquad \alpha_v = \frac{26}{9}, \qquad v(0, x) = Ae^{-x^2}.$$

Consequently:

(E4.5) *If $\nu = 0$, then $v(t, 0) = \dfrac{A}{1 - At}$ blows up at $T_{\max} = 1/A$.*

(E4.6) *If $\nu = 1$, define*

$$A_*^{(E4)} := \frac{70}{9}\, 2^{35/18}. \tag{3.3}$$

If $A > A_*^{(E4)}$, then blow-up occurs in finite time, located at $x = 0$.

Proof. Identical to the previous Kaplan functional proof: use $\phi = e^{-x^2}$ and the weight $|x|^{\alpha_v}$. □

3.3. Blow-up for the $\epsilon > 0$ cases (E4.1)–(E4.4).

Theorem 3.6 (Blow-up for $\epsilon > 0$ with quartic seed)**.** *Assume $\epsilon > 0$ and let (u, v) be the classical solution of (3.1)–(3.2). Then $T_{\max} < \infty$ and $v(t, 0) \to +\infty$ as $t \uparrow T_{\max}$ in the following regimes:*

(E4.1) *$\kappa = \nu = 0$: blow-up at $x = 0$ occurs for every $A > 0$, with $T_{\max} = 1/A$.*

(E4.3) *$\kappa = 1$, $\nu = 0$: there exist constants $A_1 > 0$ and $c_1 > 0$ (depending only on $\alpha_u = \frac{7}{3}$) such that if*

$$A \geq A_1, \qquad 0 < \epsilon \leq c_1 A^3,$$

then blow-up occurs in finite time at $x = 0$.

(E4.2), (E4.4) *If $\nu = 1$ (with $\kappa = 0$ or 1) and A is sufficiently large while ϵ is sufficiently small, then blow-up occurs in finite time at $x = 0$.*

Proof. **(E4.1).** At $x = 0$, $u(0, 0) = 0$ implies $u(t, 0) \equiv 0$, hence $v(t, 0)$ solves $V' = V^2$ and blows up at $1/A$.

(E4.3). At $x = 0$, $V' = V^2 - U^2$. Using Duhamel and Lemma 2.5 with $\alpha = \alpha_u = 7/3$ yields $(S_{\alpha_u}(t)b)(0) \leq C_{\alpha_u}\epsilon t^2$, hence $U(t) \lesssim \epsilon/A^2$ on $t \asymp 1/A$. For $\epsilon \leq c_1 A^3$ this ensures $U \ll V$ and therefore $V' \gtrsim V^2$, giving blow-up.

(E4.2), (E4.4). When $\nu = 1$, apply the Kaplan functional for v as in (E4.6), with an additional nonpositive term $-\int u^2\phi|x|^{\alpha_v}dx$. On $t \asymp 1/A$, the quartic-origin control keeps u perturbatively small, so the Riccati mechanism closes for A large and ϵ small. □

Remark 3.7 (Quadratic seed also works)**.** *Replacing $b(x) = \epsilon x^4 e^{-x^2}$ by $b(x) = \epsilon x^2 e^{-x^2}$ also yields blow-up, at the cost of replacing the t^2-gain in Lemma 2.5 by a t-gain.*

4. Summary

We analyzed finite-time blow-up for two one-dimensional convection-free nonlinear diffusion systems with Bessel-type diffusion. The first family (B3) is motivated by the Constantin–Lax–Majda (CLM) model for vortex stretching: its inviscid base subcase (B3.1) coincides with CLM under an explicit identification, and the complex variable $z = v + i\beta$ yields the reduction $z_t = z^2$. The second family (E4) replaces the $v^2 - u$ coupling in (B3) by $v^2 - u^2$.

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